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Model Practical lab

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Networks

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- 4) Design and configuration of RING Topologies using Packet Tracer.
- 33) Implementing the applications using Tcp file transfer in Java/c.

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100

- 4) Aim :- To Design and configuration of RING topologies using Packet tracer and hence to transmit data between the devices connected using Ring topology.

Software / Apparatus Required :- Packet Tracer / End devices, Hubs, connectors

Steps for building topology :-

step 1 :- start packet Tracer

step 2 :- choosing devices and connections

step 3 :- Building the Topology - Adding Hosts.

Simple click on the End Devices

Single click on the Generic Host

Move the cursor into topology area.

Single click in the topology area and it copies the devices

Step 4 - Building the Topology - connecting the Hosts to switches.

Select a switch, by clicking once a switches and once on a 2950-24 switch.

Add the switch by moving the plus sign "+".

Steps - connect PCs to switch by first choosing connections.

click once on the copper straight through cable

click once on PC2

Choose Fast Ethernet

Drag the cursor to switch 0

click once on switch 0

Step 6 - configuration IP Addresses and subnet Masks on the Hosts.

To start communication between the hosts IP Addresses and subnet Masks has to be configured on a device. click once on PC0. choose the configuration tab and click on automatically

on the node. select desktop option and then command Prompt. once the window pops up, Ping the Ip address of the device at which node is connected. Ping statistics will be displayed.

Result - Thus the Ring topology is implemented with Packet Tracer simulation Tool.

33) Aim - To implement the applications using TCP file transfer in Java.

Algorithm -

- 1) start the program
- 2) Declare the variables and structure required
- 3) A socket is created and the connect function is executed.
- 4) The file is opened.
- 5) The data from the file is read and sent to the server
- 6) The socket is closed
- 7) The Program is stopped.

Steps to implement a TCP file transfer:

- 1) Set up the TCP client-server connection.
 - create a server socket using 'socket()' function
 - Bind the server socket to a specific IP address and port using the 'bind()' function
 - create a client socket using socket function.
- 2) Server side implementation.
 - Accept the client connections using 'accept()' function
 - open the file to be transferred in binary mode.
 - Read the contents of the file in chunks and send the over TCP connections.
 - close the file using 'fclose()'
- 3) client-side implementation.
 - Receive the file data from the server using 'recv()' function on the client side.
 - write the received data to a file on the client side using 'fwrite()'
 - close the file using 'fclose()'.

Result: Thus the applications using TCP file transfer in Java is completed successfully

output

From PC0, execute:

C:\> ping 192.168.1.2

Reply from 192.168.1.2 : bytes = 32 time < 1ms TTL = 128

Reply from 192.168.1.3 : bytes = 32 time < 1ms TTL = 128

Reply from 192.168.1.4 : bytes = 32 time < 1ms TTL = 128

Reply from 192.168.1.1 : bytes = 32 time < 1ms TTL = 128

Ping statistics from 192.168.1.2 :

Packets : Sent = 4 Received = 4 Lost = 0
(0% loss)

output

TCP File Transfer

Server started...

waiting for client connection

client connected successfully

Enter the file name to transfer : sample.txt

File found

File transfer started..

File transfer completed successfully

server : file sent successfully

client : file received successfully